Ex) $(\log_2 x)^4 - \left[\log_{1/2} \frac{x^5}{4}\right]^2 - 20\log_2 x + 148 < 0$

$\rightarrow (\log_2 x)^4 - \left[\log_{1/2} \frac{x^5}{4}\right]^2 - 20\log_2 x + 148 < 0$

$\rightarrow$ Let $\log_2 x = t$

$\rightarrow$ Simplify $\log_{\frac{1}{2}} \frac{x^5}{4}$

$\Rightarrow \log_{\frac{1}{2}} x^5 - \log_{\frac{1}{2}} 4$

$\Rightarrow 5\log_{\frac{1}{2}} x - (-2)$

$\Rightarrow -5\log_2 x + 2$

$\rightarrow (\underset{t}{\log_2 x})^4 - [\underset{t}{-5\log_2 x} + 2]^2 - 20\underset{t}{\log_2 x} + 148 < 0$

$\rightarrow (t)^4 - [(-t) + 2]^2 - 20t + 148 < 0$

$\rightarrow t^4 - [(-t)^2 + 2(2)(-t) + (2)^2] - 20t + 148 < 0$

$\rightarrow t^4 - [25t^2 - 20t + 4] - 20t + 148 < 0$

$\rightarrow t^4 - 25t^2 + 20t - 4 - 20t + 148 < 0$

$\rightarrow t^4 - 25t^2 + 144 < 0$

Let $\log_2 x = t$, $t^2 = y$

$\Rightarrow (t^2)^2 - 25(t^2) + 144 < 0$

$\Rightarrow y^2 - 25y + 144 < 0$

$\Rightarrow y^2 - 9y - 16y + 144 < 0$

$\Rightarrow y(y-9) - 16(y-9) < 0$

$\Rightarrow (y-9)(y-16) < 0$

$\Rightarrow y \Rightarrow 9, 16$

Now

$\Rightarrow 9 < y^2 < 16$

$\Rightarrow 9 < t^2 < 16$

$\Rightarrow t^2 > 9$

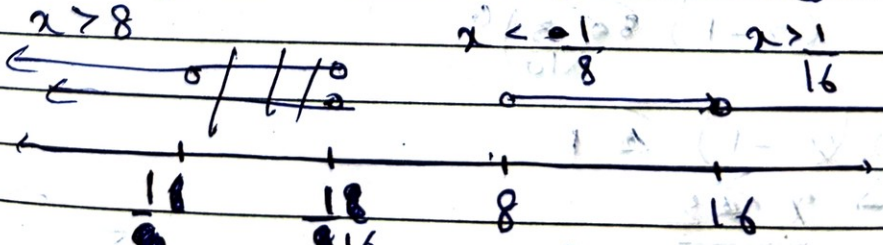
$\Rightarrow t > 3$ (or) $t < -3$ (and) $t > -4$ (or) $t < 4$

$\Rightarrow \log_2 x > 3$ (or) $\log_2 x < -3$ and $\log_2 x > -4$ (or) $\log_2 x < 4$

1) $\log_2 x > 3$ $\quad$ 2) $\log_2 x < -3$ $\quad$ (3) $\log_2 x > 4$ $\quad$ 4) $\log_2 x < 4$

$x > (2)^3$ $\quad$ $x < (2)^{-3}$ $\quad$ $x > (2)^{-4}$ $\quad$ $x < (2)^4$

$x > 8$ $\quad$ $x < \frac{1}{8}$ $\quad$ $x > \frac{1}{16}$ $\quad$ $x < 16$



$\Rightarrow \left(\frac{1}{16}, \frac{1}{8}\right) \cup (8, 16)$

Home - Work

Ex) If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, Then x lies in the interval

$\rightarrow \log_{0.3}(x-1) < \log_{0.09}(x-1)$

$\rightarrow \log_{0.3}(x-1) - \log_{0.09}(x-1) < 0$

$\rightarrow \log_{0.3}(x-1) - \log_{\frac{9}{100}}(x-1) < 0$

$\rightarrow \log_{\frac{3}{10}}(x-1) - \log_{\left(\frac{3}{10}\right)^2}(x-1) < 0$

$\rightarrow \log_{\frac{3}{10}}(x-1) - \frac{1}{2}\log_{0.3}(x-1) < 0$

$\rightarrow 2\log_{0.3}(x-1) - \log_{0.3}(x-1) < 0$

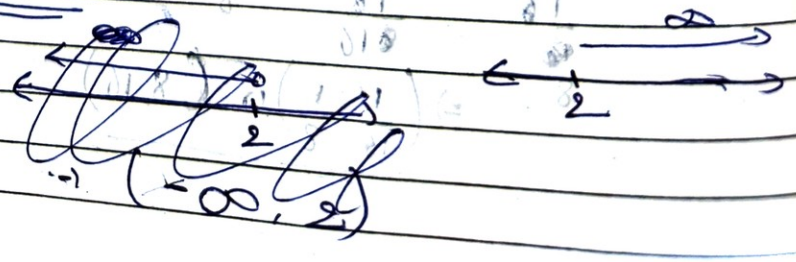
$\rightarrow \log_{0.3}(x-1) < 0$

$\rightarrow (x-1) > \left(\frac{3}{10}\right)^0$

$\rightarrow (x-1) > 1$

$\rightarrow x > 2$

$\therefore (-\infty, 2)$



$x - 1 \geq 0$

$\Rightarrow x \geq 1$



$\therefore (1, 2)$, $(2, \infty)$ final ans

x) The Solution of the inequality

$\log_{\frac{5}{8}} \left[2x^2 - x - \frac{3}{8} \right] \geq 1$ is

$\Rightarrow \log_{\frac{5}{8}} \left[2x^2 - x - \frac{3}{8} \right] > 1$

$\Rightarrow \left[2x^2 - x - \frac{3}{8} \right] \geq \frac{5}{8}$

$\Rightarrow 16x^2 - 8x - 3 \geq 5$ → forgot to change the sign of inequality

$\Rightarrow 16x^2 - 8x - 3 + 3 \geq 5$

$\Rightarrow 16x^2 - 8x - 8 \geq 0$

$\Rightarrow 8(2x^2 - x - 1) \geq 0$

$\Rightarrow 2x^2 - x - 1 \geq 0$ → changed it

$\Rightarrow 2x^2 - x - 1 \leq 0$

$\Rightarrow 2x^2 - 2x + x - 1 \leq 0$

$\Rightarrow 2x(x - 1) + 1(x - 1) \leq 0$

$\Rightarrow (x - 1)(2x + 1) \leq 0$

$\Rightarrow x = 1$ and $-\frac{1}{2}$



$\Rightarrow \left[-\frac{1}{2}, 1 \right]$

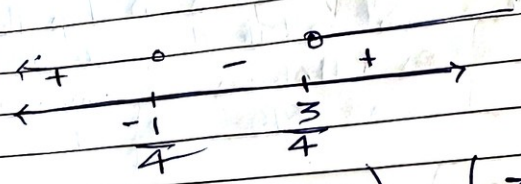
$\Rightarrow$

$2x^2 - x - \frac{3}{8} > 0$

$16x^2 - 8x - 3 > 0$

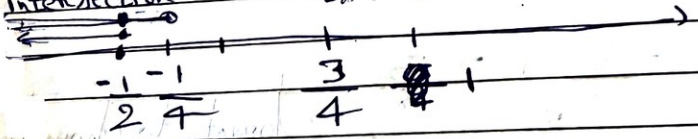
$[4x+1][4x-3] > 0$

$\rightarrow x \Rightarrow \frac{-1}{4}$ and $\frac{3}{4}$



$\Rightarrow \left(-\infty, \frac{-1}{4}\right) \cup \left(\frac{3}{4}, \infty\right)$

$\Rightarrow$ intersection



$\rightarrow \left[\frac{-1}{2}, \frac{-1}{4}\right) \cup \left(\frac{3}{4}, 1\right]$

Logarithmic Inequalities (With Variable base)

∴ $f(x) \Rightarrow \log_a x$

1) $x > 0$
2) $a > 0$
3) $a \neq 1$

⇒ base > 1 ⇒ same
⇒ base < 1 ⇒ change

∴ $\log_x \left(2x - \frac{3}{4}\right) > 2$

⇒ Two cases Whenever the base is Variable

Case-1 ⇒ $0 < x < 1$ (or) Case-2 $x > 1$

→ $\log_x \left(2x - \frac{3}{4}\right) > 2$

→ $2x - \frac{3}{4} > x^2$

→ $2x - \frac{3}{4} < x^2$

→ $8x - 3 < 4x^2$

→ $0 < 4x^2 - 8x + 3$

→ $4x^2 - 8x + 3 > 0$

→ $4x^2 - 6x - 2x + 3 > 0$

→ $4x^2 - 2x - 6x + 3 > 0$

→ $2x(2x-1) - 3(2x-1) > 0$

→ $(2x-1)(2x-3) > 0 \Rightarrow$

→ $2x - \frac{3}{4} > x^2$

→ $4x^2 < 8x - 3$

→ $4x^2 - 8x - 3 < 0$

→ $(2x-1)(2x-3) < 0$

Case -1

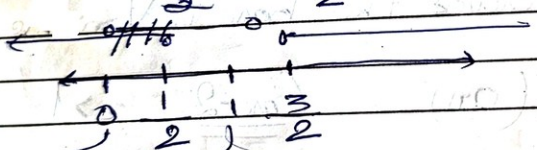
Note:-

We took a Case-1 where the "x" is in b/w $0 < x < 1$ and after solving the expression we will get an answer of "x" from the function, after that we will take intersection of the Case that we assumed and the answer of the function and we will form intervals.

$\rightarrow 0 < x < 1$

$(2x-3)$ and $(2x-1) > 0$

$x \Rightarrow \frac{3}{2}$ and $\frac{1}{2}$



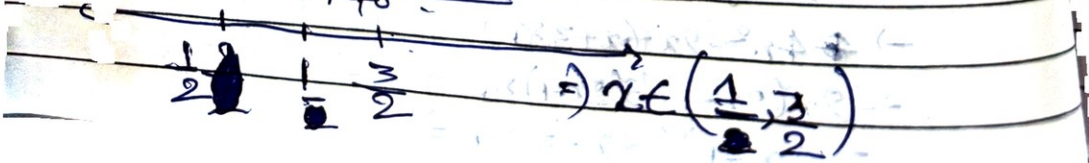
from the case $\Rightarrow x \in \left(0, \frac{1}{2}\right)$

$\Rightarrow$ Case -2

$\rightarrow$ Same as we did in Case-1

$\Rightarrow x > 1$

$(2x-3)(2x-1) < 0$



$\Rightarrow x \in \left(\frac{1}{2}, \frac{3}{2}\right)$

Note:
Then after getting answers for both Cases We Will Do the union of both of the Case.

$x \in (0, \frac{1}{2})$, $x \in (1, \frac{3}{2})$

⇒ final answer :-

$$[\text{Case}(1) \cup \text{Case}(2)] \cap \text{Domain}$$

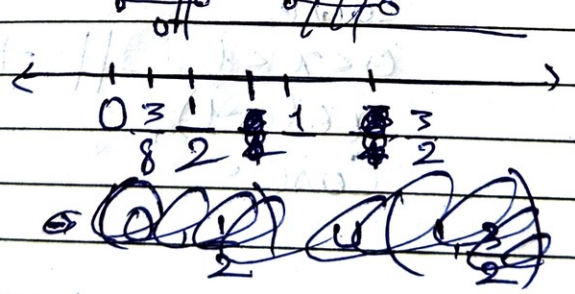
⇒ Domain,

$$\frac{2x - 3}{4} > 0$$

$$8x - 3 > 0$$

$$8x > 3$$

$$x > \frac{3}{8}$$



$$\Rightarrow \left(\frac{3}{8}, \frac{1}{2}\right) \cup \left(1, \frac{3}{2}\right)$$

Ex) $\log_x (x^3 - x^2 - 2x) < 3$

→ Case - 1

$0 < x < 1$

$\log_x (x^3 - x^2 - 2x) < 3$

$(x^3 - x^2 - 2x) < x^3$

$(x^3 - x^2 - 2x - x^3) > 0$

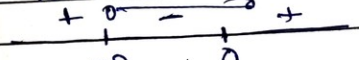
$-x^2 - 2x > 0$

$x^2 + 2x < 0$

$x(x+2) < 0$

$x \Rightarrow 0$

$x \Rightarrow -2$



$\Rightarrow (-2, 0)$ { No Solution Cause $0 < x < 1$ ~~#~~ $x \in (0, 1)$ Case }

⇒ Case - 2

→ $x > 1$

→ $x^3 - x^2 - 2x < x^3$

→ $x^3 - x^2 - 2x - x^3 < 0$

→ $-x^2 - 2x < 0$

→ $x^2 + 2x > 0$

→ $x(x+2) > 0$

→ $x \Rightarrow 0, -2$



$\Rightarrow (-\infty$

⇒ Case 1 and Case 2 ⇒ Union

⇒ $(-2,0) \cup (-\infty,-2) \cup (0,\infty)$

Domain,

$\Rightarrow x^3 - x^2 - 2x > 0$

$\rightarrow x^3 - x^2 - 2x > 0$

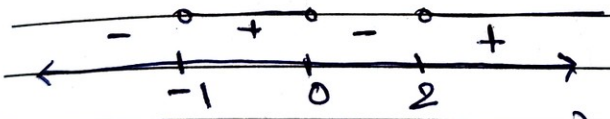
$\rightarrow x(x^2 - x - 2) > 0$

$\rightarrow x(x^2 - 2x + x - 2) > 0$
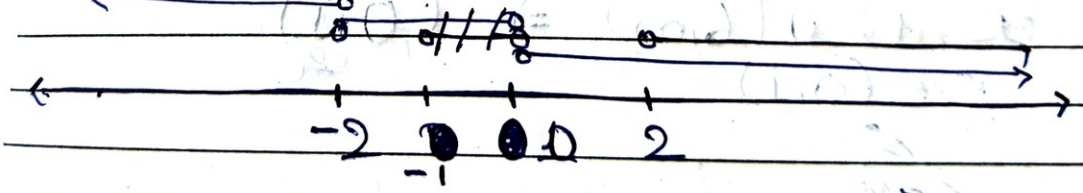
$\rightarrow x(x(x-2) + 1(x-2)) > 0$

$\rightarrow ((x-2)(x+1))x > 0$

$x \Rightarrow 0, 2, -1$



$\Rightarrow (-1,0) \cup (2,\infty) \cap (-2,0) \cup (-\infty,-2) \cup (0,\infty)$



$\rightarrow (-1,0) \cup (2,\infty)$ $\rightarrow (-1,0) \cup (2,\infty)$

$\rightarrow (-1,2)$

Ex) $\log_{2x}(x^2-5x+6) < 1$

→ Case - 1

$\log_{2x}(x^2-5x+6) < 1$

$0 < x < 1, \quad 0 < 2x < 1 \Rightarrow 0 < x < \frac{1}{2}$

$\log_{2x}(x^2-5x+6) < 1$

$(x^2-5x+6) < (2x)^1$

$x^2-5x+6-2x > 0$

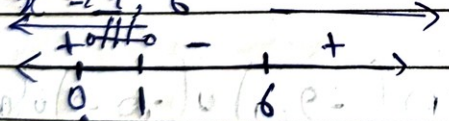
$x^2-7x+6 > 0$

$x^2-6x-x+6 > 0$

$x(x-6)-1(x-6) > 0$

$(x-1)(x-6) > 0$

$x = 1, 6$



$(-\infty, 1) \cup (6, \infty) \Rightarrow (0, \frac{1}{2})$

$x \in (0, 1)$

∈

case

Case - 2

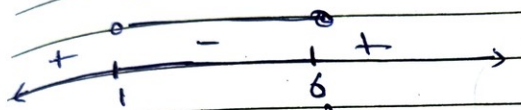
$\log_{2x}(x^2-5x+6) < 1$

$\rightarrow 2x > 1$

$\Rightarrow x^2 - 5x + 6 < 2x$

$\rightarrow x^2 - 5x + 6 - 2x < 0$

$\rightarrow (x-1)(x-6)$



$\Rightarrow (1, 6)$

$\rightarrow (1,6) \cup (0, \frac{1}{2})$

Case 1 and 2

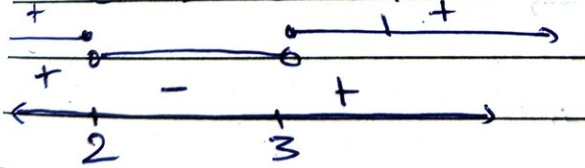
Domain :-

$x^2 - 5x + 6 \geq 0$

$x^2 - 3x - 2x + 6 \geq 0$

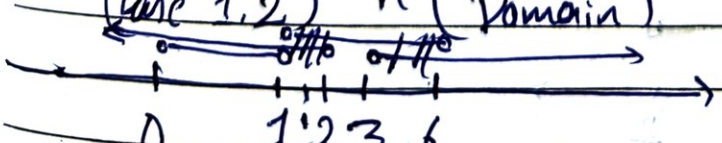
$x(x-3) - 2(x-3) \geq 0$

$(x-3)(x-2) \geq 0$



$\Rightarrow (-\infty, 2) \cup (3, \infty)$

$\rightarrow$ (Case 1,2) $\cap$ (Domain)

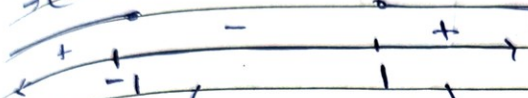


$\rightarrow (0, \frac{1}{2}) \cup (1, 2) \cup (3, 6)$

Case-2 : $x^2 > 1$

$\Rightarrow x^2 - 1 > 0$

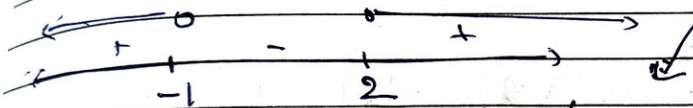
$\Rightarrow (x-1)(x+1) > 0$



$x \in (-\infty, -1) \cup (1, \infty)$

$\Rightarrow \log_{x^2}(2+x) < 1$

$(x-2)(x+1) > 0$



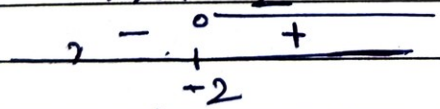
$x \in (-\infty, -1) \cup (2, \infty)$

$x \in (-\infty, -1) \cup (1, \infty) \cup (2, \infty) \cup (-1, 0) \cup (0, 1)$

Domain

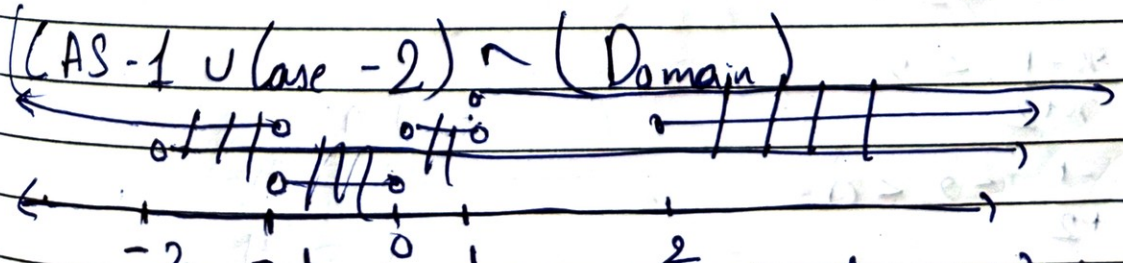
$\rightarrow 2 + x > 0$

$\rightarrow x > -2$



$x \in (-2, \infty)$

(AS-1 $\cup$ Case-2) $\cap$ (Domain)



$x \in (-2, -1) \cup (2, \infty) \cup (-1, 0) \cup (0, 1)$

Ex)

$$\log_{\frac{x+6}{3}}\left[\log_2 \frac{x-1}{x+2}\right] > 0$$

→ Base > 1 inequality don't change

→ Case-1 :

$$0 < x < 1$$

$$0 < \frac{x+6}{3} < 1$$

$$0 < x+6 < 3$$

$$0 < x < -3$$

$$-3 < x < 0 \quad , \quad x \in (-3, 0)$$

$$\log_{\frac{x+6}{3}}\left[\log_2 \frac{x-1}{x+2}\right] > 0$$

$$\log_2 \frac{x-1}{x+2} < \left(\frac{x+6}{3}\right)^0$$

$$\log_2 \frac{x-1}{x+2} < 1$$

$$\log_2 \frac{x-1}{x+2} < 1$$

$$\frac{x-1}{x+2} < 2$$

$$\frac{x-1}{x+2} - 2 < 0$$

$$\frac{x-1-2x-4}{x+2} < 0$$

Case -1

$$\left.\begin{array}{l} 0 < x < 1 \\ 0 < x+6 < 1+6 \\ 6 < x+6 < 7 \end{array}\right\}$$

$$0 < x < 1$$
$$0 < \frac{x+6}{3} < 1$$
$$0 \times 3 < x+6 < 1$$
$$0 < x+6 < 3$$
$$-6 < x < -3$$

$x \in (-6, -3)$

$$\rightarrow \log_{\frac{x+6}{3}} \left[\log_2 \frac{x-1}{x+2} \right] > 0$$

$$\rightarrow \log_2 \frac{x-1}{x+2} < 1$$

$$\rightarrow \frac{x-1}{x+2} < 2$$

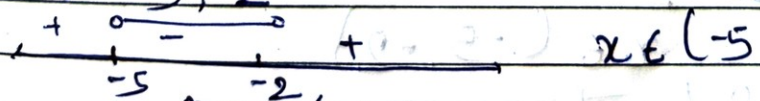
$$\rightarrow \frac{x-1}{x+2} - 2 < 0$$

$$\rightarrow \frac{x-1-2x-4}{x+2} < 0$$

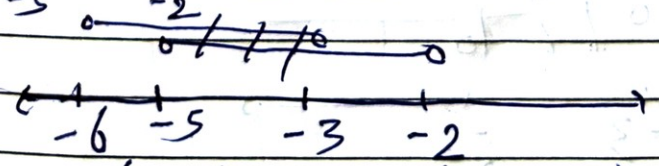
$$\rightarrow \frac{-x-5}{x+2} < 0$$

$$\rightarrow \frac{x+5}{x+2} < 0$$

$x \Rightarrow -5, -2$



$x \in (-5, -2)$



Case 1: $x \in (-5, -3)$

Case-2

$x > 1$

$\frac{x+6}{3} > 1$

$x+6 > 3$

$x > -3$, $x \in (-3, \infty)$

$\log_{\frac{x+6}{3}} \left[\log_2 \frac{x-1}{x+2}\right] > 0$

$\log_2 \frac{x-1}{x+2} > 1$

$\frac{x-1}{x+2} - 2 > 0$

$\frac{x-1-2x-4}{x+2} > 0$

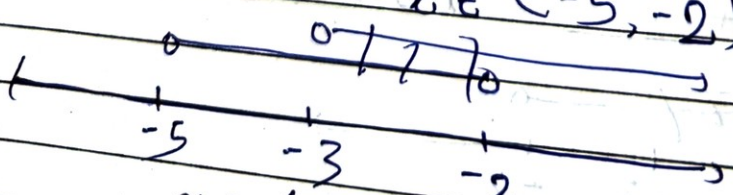
$\frac{-x-5}{x+2} > 0$

$\frac{x+5}{x+2} < 0$

$x \Rightarrow -5, -2$



$x \in (-5, -2)$



$x \in (-3, -2)$

Case 1 and 2 → $(-5, -3) \cup (-3, -2)$

Doma

$\frac{x}{x}$

x

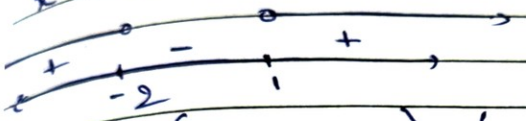
$+$

$\Rightarrow$

Domain.

$\frac{x-1}{x+2} > 0$

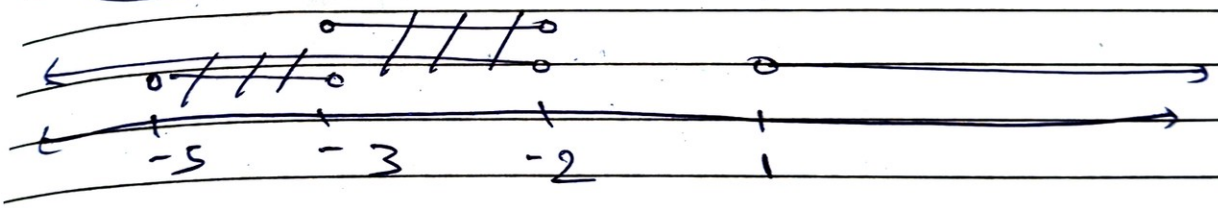
$x \Rightarrow 1, -2$



$x \in (-\infty, -2) \cup (1, \infty)$

$\Rightarrow$ (Case -1 & 2) $\cap$ (Domain)

$\Rightarrow ((-5, -3) \cup (-3, -2)) \cap ((-\infty, -2) \cup (1, \infty))$



$\rightarrow (-5, -3) \cup (-3, -2)$